

Biosynthesis of SiO₂ nanoparticles using extract of *Nerium oleander* leaves for the removal of tetracycline antibiotic

Noureddine El Messaoudi^{a,*}, Mohammed El Khomri^a, El-Houssaine Ablouh^b, Amal Bouich^c, Abdellah Lacherai^a, Amane Jada^d, Eder C. Lima^e, Farooq Sher^{f,**}

^a Laboratory of Applied Chemistry and Environment, Ibn Zohr University, Agadir, 80000, Morocco

^b Materials Science, Energy and Nanoengineering Department (MSN), Mohammed VI Polytechnic University (UM6P), Benguerir, 43150, Morocco

^c Department of Applied Physics, Institute of Design and Manufacturing (IDF), Polytechnic University of Valencia, Valencia, 46000, Spain

^d Institute of Materials Science of Mulhouse (IS2M), High Alsace University, Mulhouse, 68100, France

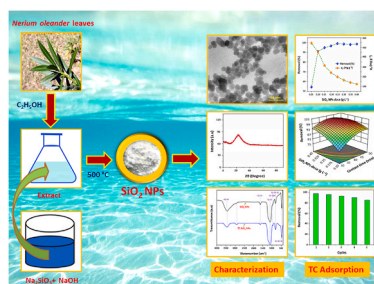
^e Institute of Chemistry, The Federal University of Rio Grande do Sul (UFRGS), Porto Alegre, RS, 91501-970, Brazil

^f Department of Engineering, School of Science and Technology, Nottingham Trent University, Nottingham, NG11 8NS, United Kingdom

HIGHLIGHTS

- SiO₂ nanoparticles were biosynthesized from an extract of *Nerium oleander* leaves.
- SiO₂ NPs is efficient adsorbent to remove tetracycline from water.
- The maximum quantity adsorbed was 552.48 mg/g.
- SiO₂ NPs have excellent reusability using 0.1 M NaOH.
- The TC removal was 99.56% using BBD-RSM in optimal conditions.

GRAPHICAL ABSTRACT



ARTICLE INFO

Handling editors:

Keywords:

Green synthesis
Tetracycline
SiO₂ nanoparticles
Nerium oleander leaves Extract
Chemical engineering
Adsorption and optimization

ABSTRACT

Tetracycline (TC) is one of the antibiotics that is found in wastewaters. TC is toxic, carcinogenic, and teratogenic. In this study, the tetracycline was removed from water by adsorption using dioxido silicon nanoparticles (SiO₂ NPs) biosynthesized from the extract of *Nerium oleander* leaves. These nanoparticles were characterized using SEM-EDX, BET-BJH, FTIR-ATR, TEM, and XRD. The influences of various factors such as pH solution, SiO₂ NPs dose, adsorption process time, initial TC concentration, and ionic strength on adsorption behaviour of TC onto SiO₂ NPs were investigated. TC adsorption on SiO₂ NPs could be well described in the pseudo-second-order kinetic model and followed the Langmuir isotherm model with a maximum adsorption capacity was 552.48 mg/g. At optimal conditions, the experimental adsorption results indicated that the SiO₂ NPs adsorbed 98.62% of TC. The removal of TC using SiO₂ NPs was 99.56% at conditions (SiO₂ NPs dose = 0.25 g/L, C₀ = 25 mg/L, and t = 40 min) based on Box–Behnken design (BBD) combined with response surface methodology (RSM) modelling. Electrostatic interaction governs the adsorption mechanism is attributed. The reusability of SiO₂ NPs was tested, and the performance adsorption was 85.36% after the five cycles. The synthesized SiO₂ NPs as promising adsorbent has a potential application for antibiotics removal from wastewaters.

* Corresponding author.

** Corresponding author.

E-mail addresses: noureddine.elmessaoudi@edu.uiz.ac.ma (N. El Messaoudi), Farooq.Sher@ntu.ac.uk (F. Sher).

1. Introduction

The excessive accumulation of toxic antibiotics in natural aquatic systems has received enormous attention due to their acute toxicity and possible carcinogenic effect (Grenni et al., 2018). Various antibiotics such as quinolones, sulfonamides, macrolides, and tetracyclines are broadly used to prevent and treat infectious diseases (Ahmad et al., 2019). Antibiotic residues, frequently found in soil, sediment, and aquatic environments, have adverse side effects as bacterial resistance changes in the microbial ecological functions (Sodhi et al., 2021). Antibiotic accumulation may also severely impair human physiological functions and have carcinogenic, teratogenic, or hormonal effects (Liu et al., 2017); their excess entrance into food chains may cause different disorders in the human body, such as the gastrointestinal system (Rashidi Nodeh et al., 2020). Therefore, controlling and handling antibiotic contaminants is necessary to have a safe environment (Yu et al., 2016).

Tetracycline (TC) is considered one of the most dangerous antibiotics due to its toxic, carcinogenic, and mutagenic effects (Scaria et al., 2021). TC is commonly used for animal husbandry and poultry industries worldwide to promote animal growth and prevent infections (Song et al., 2020; Van et al., 2020). About 60–80% of tetracyclines were subject to different natural surroundings original in or metabolized forms due to lower metabolic rates in animals and humans (Epps and Blaney, 2016; Scaria et al., 2021); due to the high solubility rate, TC is detected in different water bodies and some regions; its concentration exceeds standard environmental limits (Daghrir and Drogui, 2013). Several approaches such as photocatalysis (Zhu et al., 2013), flocculation (Fu et al., 2015), biological treatment (Belkheiri et al., 2011), electrochemical (Wang et al., 2021), adsorption (Sharma et al., 2020), and reverse osmosis (Rostam and Taghizadeh, 2020) have been used to remove tetracycline from wastewaters. The adsorption process is a competitive and practical method for removing TC from wastewater due to high efficiency, low energy demand, and the possibility of reusing adsorbent materials (El Khomri et al., 2020; El Messaoudi et al., 2016a).

Consequently, a wide range of adsorbent materials (clays, chitosan, coated silica gel, zeolite ionic liquids, metal oxides, nanocomposites, nanomaterials, industrial and agricultural wastes) have already been tested for the recovery of TC contained in wastewater (Chang et al., 2012; Gao et al., 2012; Hao et al., 2021; Maged et al., 2020; Zhou et al., 2017). Recently, conducting nanoparticles such as ZrO_2 (Debnath et al., 2020), NiFe (Ravikumar et al., 2019), CeO_2 (Nurhasanah et al., 2020), La_2S_3 (Rashidi Nodeh et al., 2020), and ZnO (Bembibre et al., 2022) have been categorized as efficient adsorbents for TC removal from wastewaters due to high stability, porous nature, good adsorption aptitude, simple doping/dedoping, and ion exchange properties. Many methods have been used to synthesize the SiO_2 nanoparticles in the literature, such as solid-state, sonochemical (Masjedi-Arani et al., 2015), hydrothermal (Potapov et al., 2020), sol-gel (Dubey et al., 2015), and ultrasonic-assisted approaches (Ullah et al., 2019).

In this study, we used SiO_2 nanoparticles to remove TC from the aqueous solution because they are an excellent adsorbent to remove dyes and metals and a photocatalyst to treat wastewaters (Hosseini et al., 2018; Sharma et al., 2021). These nanoparticles are biosynthesized from the extract of *Nerium oleander* leaves. The biosynthesis of nanoparticles from the extract of agricultural solid wastes is regarded as a simple, rapid, cost-effective, and eco-friendly method for creating nanostructured materials such as metals, metal chalcogenides, and bimetal oxide (Das et al., 2018). In this work, we used the extract of *Nerium oleander* leaves for biosynthesis of SiO_2 nanoparticles. We chose this agricultural solid waste because its extract contains a significant amount of minerals, lost cost materials, and is eco-friendly and abundant in Morocco (Martín et al., 2018; Sebeia et al., 2019).

The performance of SiO_2 NPs was evaluated in this study by removing tetracycline from the solution aqueous. Besides, characterization techniques were practiced for investigating its structure and

properties, such as scanning electron microscope coupled with energy-dispersive X-ray (SEM-EDX), Brunauer–Emmett–Teller and Barrett–Joyner–Halenda (BET–BJH), Fourier transform infrared spectroscopy coupled with attenuated total reflectance (FTIR–ATR), transition electron microscope (TEM), X-ray diffraction (XRD), and point of zero charge (PZC). The present work will substantially impact as it will add unique knowledge on the adsorption potential of SiO_2 NPs synthesized from the extract of *Nerium oleander* leaves. In this current study, the SiO_2 NPs were synthesized from the *Nerium oleander* leaves extract. Different influencing experimental parameters such as pH, SiO_2 NPs dose, reaction time, TC concentration, and ionic strength were investigated. Kinetic and equilibrium models evaluated the adsorption of TC on SiO_2 NPs. The removal of TC optimized using Box–Behnken design (BBD) combined with response surface methodology (RSM). The adsorption mechanism of TC molecules on the SiO_2 NPs surface was proposed. The reusability of SiO_2 NPs was evaluated.

2. Experimental

2.1. Materials

Nerium oleander leaves were collected in Tinghir (South-East Morocco). The silicate of sodium (Na_2SiO_3), tetracycline (antibiotic, $\text{C}_{22}\text{H}_{24}\text{N}_2\text{O}_8$, MW = 444.435 g/mol), $\text{C}_2\text{H}_5\text{OH}$, HCl, and NaOH were purchased from Sigma-Aldrich. The distilled and deionized waters were used through experiments.

2.2. Biosynthesis of SiO_2 NPs

10 g of *Nerium oleander* leave powder were added in 100 mL of $\text{C}_2\text{H}_5\text{OH}$ and stirred for 3 h. After that, the solution was filtered using filter paper then the above mixture was centrifuged for obtained a clear extract of *Nerium oleander* leaves. 2 g Na_2SiO_3 and 50 mL of extract and 5 mL of NaOH (0.5 M) were mixed and stirred for 4 h (Dobrucka and Długaszewska, 2016). After precipitation, the mixture was filtered and centrifuged for separated the liquid and the precipitate (SiO_2 NPs). The residue obtained was washed with deionized water. Finally, The SiO_2 nanoparticles were over-dried (80 °C) for 24 h and calcined in the furnace at 500 °C for 3 h (Shaligram et al., 2009).

2.3. Characterization

The physicochemical properties of SiO_2 NPs were examined using different characterization techniques. The morphology and microstructures of SiO_2 NPs were examined by SEM-EDX analysis (JEOL, JSM-IT200) and TEM analysis (Philips CM-30). BET and BJH methods (Belsorp Mini II) were used to determine the surface area, total pore volume, and diameter pore of SiO_2 NPs. The chemical bond characteristics of SiO_2 NPs before and after TC adsorption was acquired by FTIR-ATR analysis (Jasco 4100). The SiO_2 NPs crystal structures were evaluated using XRD analysis recorded on a 6100-Shimadzu. The PZC of adsorbent was determined using the method reported by Fiol and Villarescu (2009).

2.4. Batch adsorption experiments

The TC adsorption onto SiO_2 NPs nanoparticles was conducted in batch mode using 12.5 mg of SiO_2 NPs in 50 mL of TC solutions with concentrations varied from 50 to 200 mg/L at 23 ± 1 °C. The influence of pH solution on adsorption was assessed and ranged from 3 to 11 and was adjusted by 0.01 M HCl acid or 0.01 M NaOH. The adsorbent dose, kinetic reaction, and ionic strength were changed from 0.05 to 0.4 g/L from 5 to 120 min and from 0 to 0.4 M, respectively. The separation of the solid-liquid phases was performed by centrifuging. The residual concentration was determined using a UV/Vis spectrophotometer (2300/Techcomp) at 376 nm as λ_{max} of TC. The quantity adsorbed q_e

Table 1

Design of experiment runs and corresponding responses for TC removal efficiency by SiO₂ NPs.

Variables	Codes			
	−1	0	1	
A = TC concentration (mg/L)	25	50	75	
B=Contact time (min)	30	40	50	
C=SiO ₂ NPs dose (g/L)	0.1	0.25	0.4	
Run	A (mg/L)	B (min)	C (g/L)	R (%)
1	25	30	0.1	98.05
2	75	30	0.1	77.11
3	25	50	0.1	98.87
4	75	50	0.1	90.11
5	25	30	0.4	99.23
6	75	30	0.4	78.13
7	25	50	0.4	99.06
8	75	50	0.4	93.55
9	25	40	0.25	99.56
10	75	40	0.25	90.57
11	50	30	0.25	89.45
12	50	50	0.25	98.66
13	50	40	0.1	90.11
14	50	40	0.4	98.65
15	50	40	0.25	98.44
16	50	40	0.25	98.71
17	50	40	0.25	98.24
18	50	40	0.25	98.53
19	50	40	0.25	98.61
20	50	40	0.25	98.72
21	50	40	0.25	98.23

(mg/g) and the and TC removal efficiency (%) were obtained using following Eqs. (1) and (2) respectively (El Messaoudi et al., 2016b):

$$q_e = \frac{(C_0 - C_e) \times V}{w} \quad (1)$$

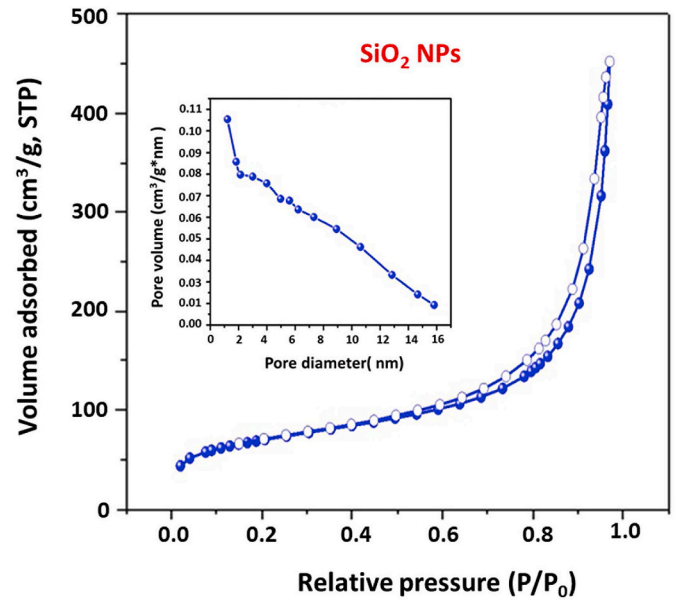


Fig. 2. N₂ adsorption/desorption isotherm curve and pore size distribution of TC-SiO₂ NPs.

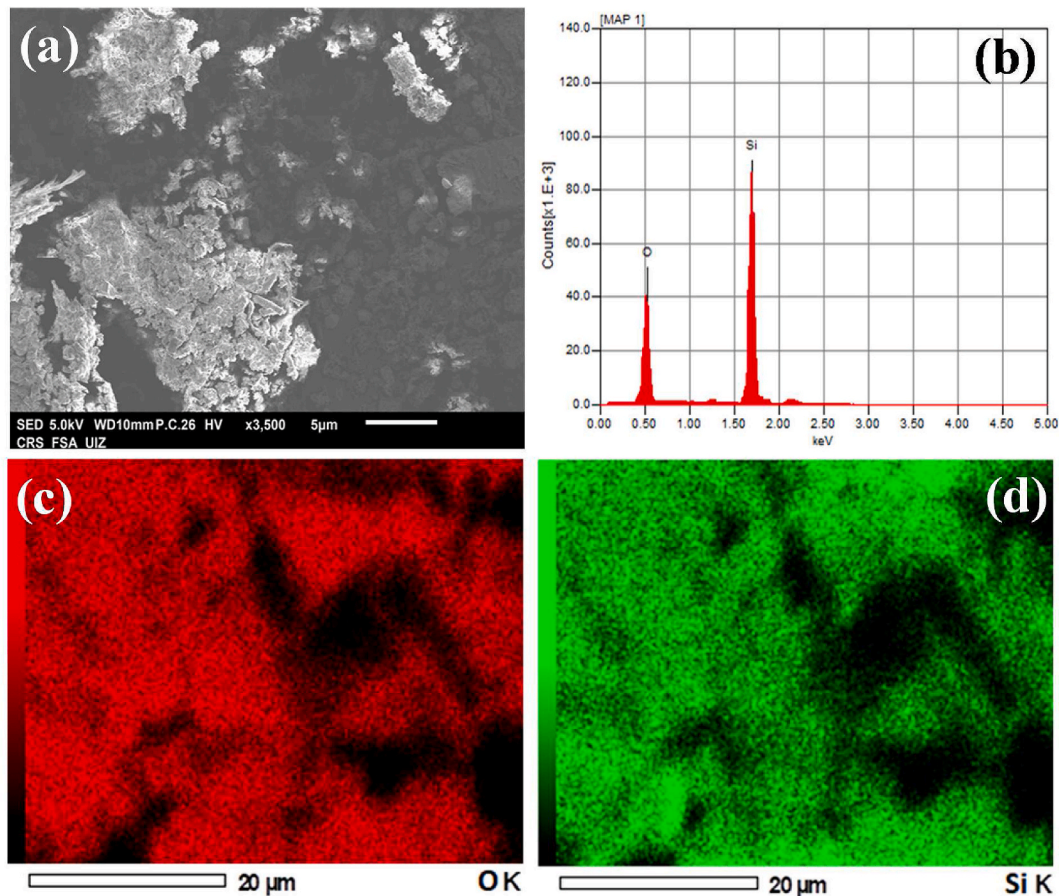


Fig. 1. SEM image; (a) SiO₂ NPs, (b) Elemental analysis, (c) Mapping image of SiO₂ NPs: oxygen and (d) Mapping image of silicon.

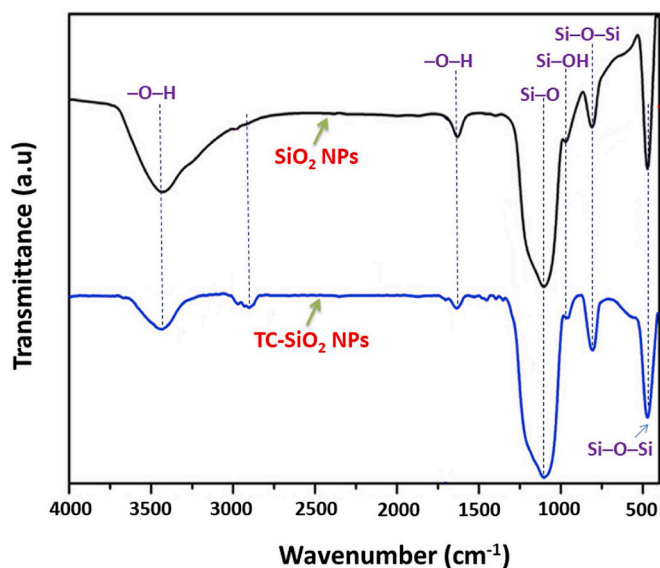


Fig. 3. FTIR spectra of SiO₂ NPs and TC-SiO₂ NPs (after adsorption).

$$\text{Percentage Removal}(\%) = \frac{(C_0 - C_e)}{C_0} \times 100 \quad (2)$$

In equations (1) and (2), C_0 (mg/L) and C_e (mg/L) denote the concentrations of TC before and after adsorption, respectively. The w (g) and V (L) represents the weight of SiO₂ NPs and the volume of the reaction solution, respectively (Bentahar et al., 2018; El Messaoudi et al., 2017).

2.5. Experimental design

BBD used a static method to design experimental parameters influencing the adsorption of TC on SiO₂ NPs using design-expert software (version 12.0.3). TC concentration (A), reaction time (B), and SiO₂ NPs dose (C) as factors have significant effects on the TC adsorption on SiO₂ NPs at 23 ± 1 °C and pH = 5. Design of experiment runs and corresponding responses for TC removal efficiency by SiO₂ NPs are summarized in Table 1. A three-factors and levels (−1, 0, and 1) were applied to 21 experiments. The TC removal efficiency R (%) was expressed using the quadratic polynomial model was formalized in Eq. (3) (Jawad et al., 2020):

$$R(\%) = \sum_{i=1}^3 \delta_i X_i + \sum_{i=1}^3 \delta_{ii} X_i^2 + \sum_{i < j} \delta_{ij} X_i X_j + \delta_0 \quad (3)$$

where δ_0 denotes constant-coefficient, δ_i is attributed to the direct effect, δ_{ii} corresponds to higher-order effect, and δ_{ij} denote reciprocal effect.

3. Results and discussion

3.1. Material characterization

The microstructures of synthesized nanoparticles were analyzed using SEM coupled with EDX. The results obtained are shown in Fig. 1. According to Fig. 1(a), The SEM image of SiO₂ NPs showed that the particles are agglomerated and have poor dispersion. Fig. 1(b) represents the EDX spectrum and elemental analysis of SiO₂ NPs. The synthesis of SiO₂ NPs confirmed by the presence of O (53.82%) and Si (46.18%) (Dubey et al., 2015). Fig. 1(c) and (d) indicate the uniform distribution of O and Si was illustrated by EDX elemental mapping.

Fig. 2 illustrates the N₂ isotherms adsorption-desorption and average diameter distribution for SiO₂ NPs nanospheres. By means, BET and BJH methods, the obtained middle surface area, pore diameter, and total pore volume were 583.46 m²/g, 3.46 nm, and 0.27 cm³/g, respectively, confirmed the porosity of SiO₂ NPs. The FTIR spectra of SiO₂ NPs before and after the adsorption of TC (TC-SiO₂) are depicted in Fig. 3. On the spectrum of SiO₂ NPs, the broad bands at 3452 cm^{−1} and 1634 cm^{−1} correspond to OH stretching vibration and absorbed water molecule, respectively (El Messaoudi et al., 2016a; Niksefat et al., 2014).

The peaks at 1083 cm^{−1}, 953 cm^{−1} and 806 cm^{−1} and 457 cm^{−1} were ascribed to the Si–O stretching vibration, Si–OH stretching vibration and Si–O–Si symmetric stretching (Rafigh and Heydarinasab, 2017), and Si–O–Si bending, respectively (Yue et al., 2019), which confirms successful synthesis of SiO₂ nanospheres. The spectrum of TC-SiO₂ NPs shows small changes that demonstrate the TC adsorption on the surface of SiO₂ NPs. Fig. 4(a) shows the nanoparticles of SiO₂ NPs using a TEM image. According to this figure, the synthesized SiO₂ NPs particles were found to be spherical in structure (Dubey et al., 2015; Nita et al., 2019). The XRD characterization results of SiO₂ NPs are provided in Fig. 4(b). A broad peak in the range of 20–30° is corresponded to the crystallite of SiO₂ particles (Rafigh and Heydarinasab, 2017), indicating the successful synthesis of SiO₂ NPs.

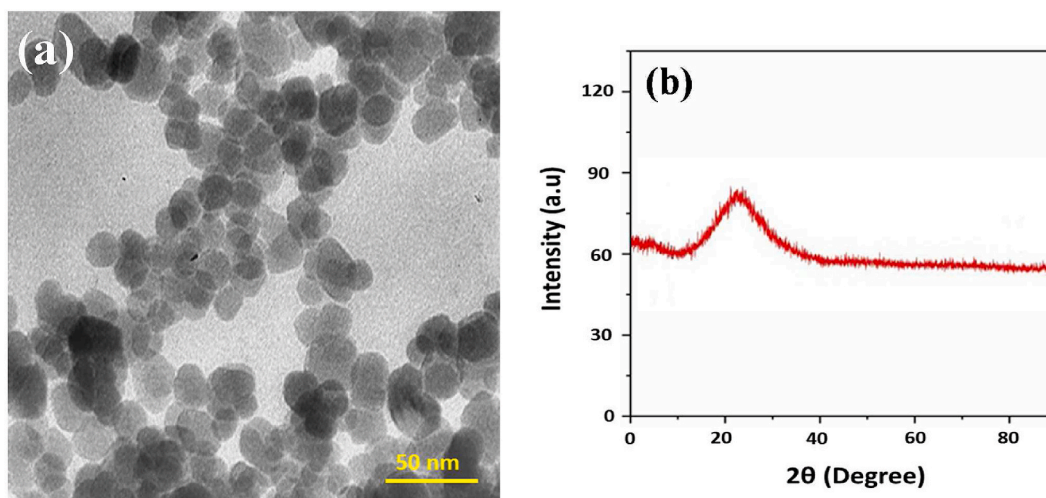


Fig. 4. SiO₂ NPs (a) TEM image and (b) XRD spectrum.

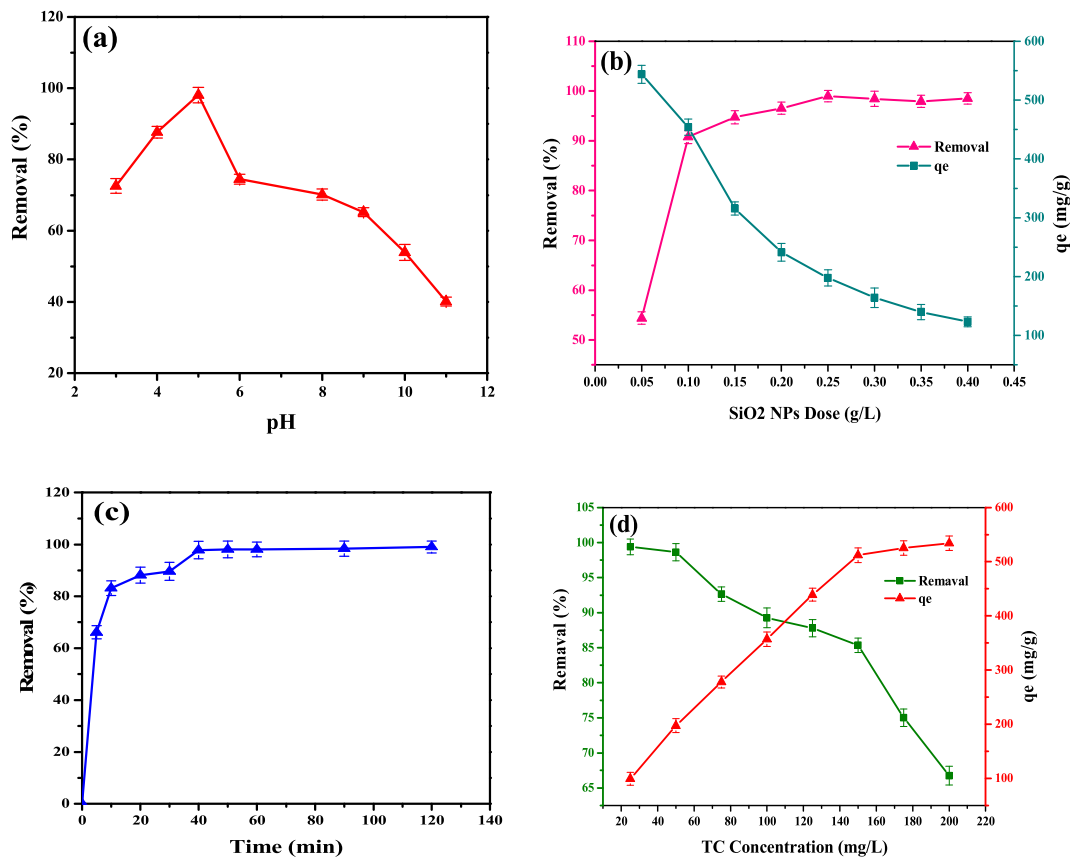


Fig. 5. Removal percentage using SiO₂ NPs; (a) Effects of pH, (b) Effect of SiO₂ NPs dose, (c) Effect of reaction time and (d) Effect of initial concentration on TC adsorption.

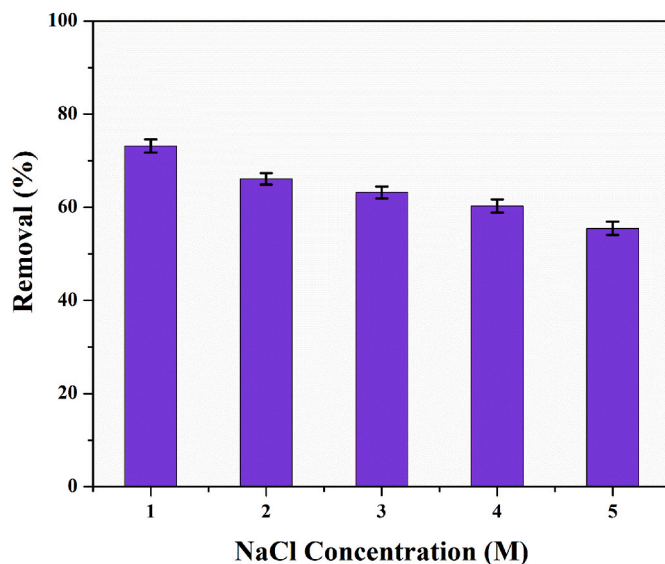


Fig. 6. Effect of ionic strength on TC adsorption using SiO₂ NPs.

3.2. Adsorption study

3.2.1. Effect of solution pH

Fig. 5(a) presents the influence of pH on the TC adsorption at pH values ranging from 3 to 11. This experiment attempted 10 mg of SiO₂ NPs in 50 mL of the solution TC (50 mg/L) at $T = 23 \pm 1^\circ\text{C}$ for 120 min. As Fig. 5(a) shows, the highest removal of TC was 98.07% at pH = 5.

Table 2

Kinetic model parameters for the TC adsorption onto SiO₂ NPs.

Model and its equation	Parameter	Value	
PFO	$q_{e,exp}$ (mg/g)	197.98	
	$q_{e,cal}$ (mg/g)	51.03	
	K_{PFO} (1/min)	0.0052	
	R^2	0.8285	
$Log(q_e - q_i) = Log(q_e - \frac{K_{PFO}}{2.303} t)$ (4)			
PSO	$q_{e,cal}$ (mg/g)	204.08	
	K_{PSO} (g/mg.min)	0.0019	
	R^2	0.9996	
$\frac{t}{q_t} = \frac{1}{K_{PSO}q_e^2} + \frac{1}{q_e} t$ (5)			
IPD	1st linear portion	K_{IPD1} (mg/g.min ^{1/2})	16.0461
		C_1 (mg/g)	35.28
		R^2	0.9811
	2nd linear portion	K_{IPD2} (mg/g.min ^{1/2})	10.1487
		C_2 (mg/g)	49.76
		R^2	0.9045
	3rd linear portion	K_{IPD3} (mg/g.min ^{1/2})	3.1460
		C_3 (mg/g)	62.75
		R^2	0.9733
	$q_t = K_{IPD}t^{1/2} + C$ (6)		

Notation: K_{PFO} (1/min) = PFO rate constant, K_{PSO} (g/mg/min) = PSO rate constant, K_{IPD} (mg/g.min^{1/2}) = IPD rate constant, C (mg/g) = Constant for any experiment, q_t (mg/g) = Amount adsorbed of TC at time t , and q_e (mg/g) = Amount adsorbed of TC at equilibrium (Graaf et al., 1990; Ho and McKay, 1998; Simonin, 2016).

Similar results were obtained by Debnath et al. (2020) and Rashidi Nodeh et al. (2020). This increase of TC removal in the acidic medium can be explained by the charge positive of SiO₂ NPs and the charge negative of TC (TCH⁻, TC⁻) (Li et al., 2010; Mohammed and Kareem, 2019). As Fig. 10(a) depicted, PZC of SiO₂ NPs was 8.2. The quantity adsorbed was 196.17 mg/g at pH = 5. The dominance of charge positive

Table 3Isotherm model parameters for the TC adsorption onto SiO₂ NPs.

Model and its equation	Parameter	Value
Langmuir $\frac{C_e}{q_e} = \frac{1}{Q_m K_L} + \frac{C_e}{Q_m} \quad (7)$	Q_m (mg/g)	552.48
	K_L (L/mg)	0.3175
	R^2	0.9931
Freundlich $\ln q_e = \ln K_F + \frac{\ln C_e}{n} \quad (8)$	K_F (mg/g)	188.7644
	n	03.5997
	R^2	0.9650
Temkin $q_e = B \ln K_T + B \ln C_e \quad (9)$	K_T (L/g)	18.3610
	B	75.6710
	R^2	0.9392

Notation: C_e (mg/L) = Equilibrium concentration of TC, Q_m (mg/g) = Mono-layer (maximum) adsorption capacity, K_L (L/mg) = Langmuir rate constant, K_F (mg/g) = Freundlich rate constant, n = Heterogeneity factor, K_T (L/g) = Temkin rate constant, and B = constant related to the heat adsorption (Freundlich, 1907; Johnson and Arnold, 1995; Langmuir, 1918).

of SiO₂ NPs at pH < PZC (adsorption of TC was favourable) and negatively charged when pH > PZC (adsorption of TC was unfavourable) (Song et al., 2020).

3.2.2. Effect of SiO₂ NPs dose

A TC concentration of 50 mg/L with a pH = 5 at T = 23 ± 1 °C for 120 min, the effect of SiO₂ NPs dose (0.05–0.4 g/L) on TC adsorption was studied. Fig. 5(b) shows the results obtained. The TC removal increased from 54.38 to 98.92% by increasing the SiO₂ NPs dose from 0.05 to 0.25 g/L, while the quantity adsorbed decreased from 543.88 to 197.95 mg/g. Results imply that the number of active adsorption sites for TC adsorption corresponds to the applied dose, prompting higher removal efficiency (Zhang et al., 2019). After equilibrium between the adsorbent and antibiotic solution, the removal percentage remains consistent at higher dosages (>0.25 g/L) (Jin et al., 2019). The optimum adsorbent dosage was considered 0.25 g/L to reach maximum TC removal efficiency, respectively.

3.2.3. Effect of contact time

The influence of the contact time on TC adsorption using SiO₂ NPs displayed in Fig. 5(a). The contact time ranged from 5 to 120 min, whereas other parameters were kept constant (SiO₂ NPs = 0.25 g/L, TC concentration = 50 mg/L, pH = 5, and T = 23 ± 1 °C). The TC adsorption was fast at first 30 min, which may be attributed to many sites accessible on the surface of the SiO₂ NPs in the initial phase (Ahmad et al., 2019). The equilibrium time was found to be 40 min. After equilibrium adsorption, the active sites were occupied by the TC molecule/ions. Therefore the adsorption rate became consistent (Zhou et al., 2020). Experimental data showed that stability was achieved in 40 min with an adsorption capacity of TC was 195.97 mg/g.

Table 4Comparison of the adsorption of SiO₂ NPs for TC with other adsorbents reported in the literature.

Adsorbent	Ad (mg/L)	T (°C)	t (min)	C ₀ (mg/L)	pH	Q _m (mg/g)	Reference
ZrO ₂ NPs	0.2	–	15	25–150	6	526.32	Debnath et al. (2020)
AgO/MgO/FeO@Si ₃ N ₄	–	30	90	30–100	8	172.41	Sharma et al. (2020)
ACCS	02.5	–	20	100–700	5	38.30	Song et al. (2020)
ZVI@ACCS	02.5	–	20	100–700	5	78.30	Song et al. (2020)
NiFe ₂ O ₄ @CDs	10	50	1440	25–100	8	591.72	Liu et al. (2017)
Pristine MoS ₂	0.4	35	2400	50–500	6	409.84	Li et al. (2021)
NiFe NPs	0.3	–	90	20–80	7	61.00	Ravikumar et al. (2019)
La ₂ S ₃ NPs	1	25	90	10–300	5	56.81	(Rashidi Nodeh et al., 2020)
SiO ₂ NPs	0.25	23	40	25–200	5	552.48	Current study

Notation: Ad = Adsorbent dose, T = Temperature, t = Time, C₀ = Initial TC concentration.

Table 5

ANOVA analysis and regression coefficients for the designed statistical model.

Source	Sun Squares	Df	Mean Square	F-value	p-value
Model	1221.20	9	135.69	162.55	<0.0001
A-TC concentration	639.35	1	639.35	765.93	<0.0001
B -Contact time	195.12	1	195.12	233.75	<0.0001
C- SiO ₂ NPs dose	10.72	1	10.72	12.85	0.0043
AB	110.78	1	110.78	132.71	<0.0001
AC	03.24	1	03.24	03.88	0.0746
BC	0.0406	1	0.0406	0.0487	0.8295
A ²	191.81	1	191.81	229.78	<0.0001
B ²	83.37	1	83.37	99.88	<0.0001
C ²	06.39	1	06.39	07.66	0.0183
Residual	09.18	11	0.8347		
Lack of Fit	08.85	5	01.77	31.76	0.0003
Pure Error	0.3343	6	0.0557		
Core total	1230.38	20			
Model statistics	R ²		Adjusted R ²	Predicted R ²	Adequate precision
	0.9925		0.9864	0.9449	37.9356

3.2.4. Effect of initial TC concentration

The influence of TC concentration (25–200 mg/L) on its retention using SiO₂ NPs was studied with a fixed pH = 5 and 0.25 g/L of SiO₂ NPs dose for 40 min. As illustrated in Fig. 5(d), by increasing TC concentration from 25 to 150 mg/L, the TC adsorption capacity progressively increased from 99.40 to 512.06 mg/g, while the removal of TC decreased from 99.40 to 85.34%. This increases TC adsorption capacity due to the occupation of all available sites on the surface of SiO₂ nanoparticles by TC molecules (Debnath et al., 2020). After 150 mg/L, a plateau was not achieved in the adsorption capacity, suggesting active sites are still available and no saturation occurred. The phenomena were similar to the adsorption research of tetracycline reported previously (Ahmad et al., 2019; Ravikumar et al., 2019).

3.2.5. Effect of ionic strength

The electrolyte (NaCl) concentration in the aqueous solution significantly affects TC adsorption onto SiO₂ NPs. As reported in Fig. 6, at SiO₂ NPs = 0.25 g/L, TC concentration = 50 mg/L, pH = 7, and T = 23 ± 1 °C, the removal efficiency of SiO₂ NPs for TC decreased from 73.15% (0 M NaCl) to 55.48% (0.4 M NaCl) by increasing the electrolyte concentrations from 0 to 0.4 M NaCl. By increasing the ionic strength, Cl[−] competes with negatively charged TC (TCH[−], TC[−]) species for adsorption onto SiO₂ NPs surface site with positively charged at pH = 7 (Zhang et al., 2019). This result is in line with another study that reported a decline in TC adsorption onto clay surface sites, negatively charged by increasing electrolyte concentration in the solution (Parolo et al., 2008). In their study, by increasing Na⁺ ion concentration at pH = 4 of the solution, the competence for occupying surface sites between Na⁺ ion and positively charged TC (TCH⁺) increased (Yang et al., 2011). The reduction in TC adsorption onto the surface of clay increased in a higher

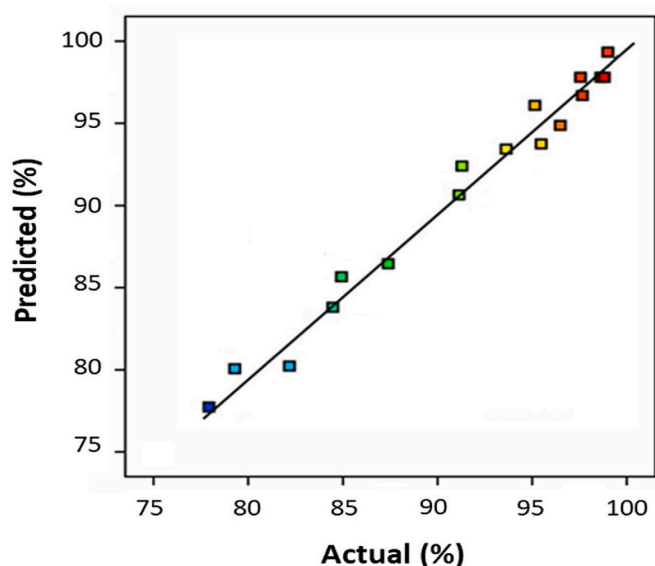


Fig. 7. Actual versus predicted for TC adsorption on SiO₂ NPs.

concentration of Na⁺ (Parolo et al., 2008).

3.3. Adsorption kinetics

The obtained experimental data were evaluated by using the pseudo-first-order (PFO) (Simonin, 2016), pseudo-second-order (PSO) (Ho and

McKay, 1998), and intraparticle diffusion (IPD) (Graaf et al., 1990) kinetic models and their linear forms are given using the following Eqs. (4)–(6) as represented in Table 2, respectively. Table 2 also shows the parameters for linear fitting. Furthermore, the PSO model was fitted to data experimental based on R² (correlation coefficient). It can be found that R² values are very near to 1, and $q_{e,exp}$ values are also closer to $q_{e,cal}$ values for the PSO. This model speculates that adsorption pursues a second-order mechanism (El Messaoudi et al., 2021b).

3.4. Adsorption isotherm

The TC adsorption on SiO₂ NPs was the Langmuir (1918), Freundlich (1907), and Temkin (Johnson and Arnold, 1995) isotherm models. The linear forms of these isotherms are expressed on Eqs. (7)–(9), as represented in Table 3. The parameters of linear fitting are listed in Table 3. Based on regression coefficients R² (0.9931, 0.9650, and 0.9392), the Langmuir model best described the TC adsorption on SiO₂ NPs. Langmuir model suggests the dye adsorption occurs as a monolayer from the homogeneous surface of the adsorbent (Bentahar et al., 2017; El Messaoudi et al., 2021a). A similar observation was found in other studies, showing successful adsorption isotherm data using Langmuir isotherm compared to Freundlich and Temkin isotherm models (Li et al., 2021; Mohammed et al., 2020). The maximum adsorption capacity Q_m of SiO₂ NPs for TC adsorption was 552.48 mg/g. The comparison of the adsorption of SiO₂ NPs to remove TC from aqueous solution with other adsorbents reported in the literature is summarized in Table 4. Based on the results presented in this table, that the SiO₂ NPs exhibit high adsorption of TC compared with adsorbents. Therefore, SiO₂ NPs are a suitable adsorbent for the removal of antibiotic molecules from

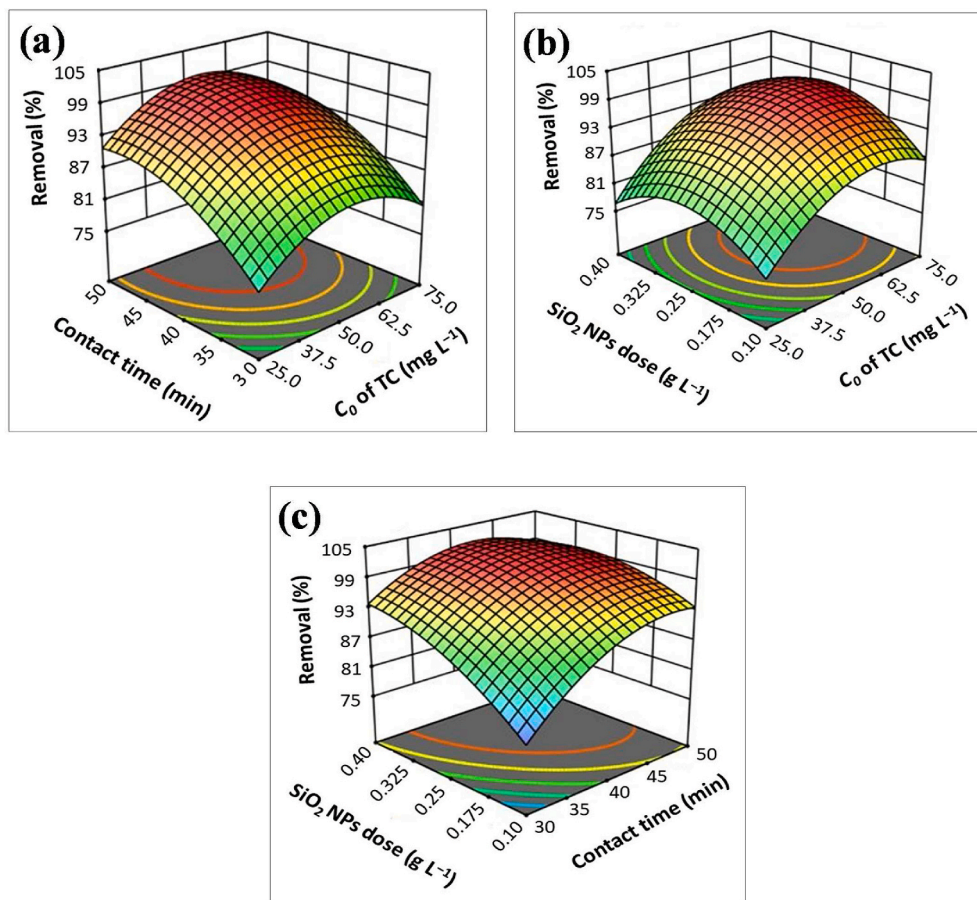


Fig. 8. Surface response plots of TC removal percentage; (a) Initial TC concentration with contact time, (b) SiO₂ NPs dose with initial TC concentration and (c) SiO₂ NPs dose with contact time.

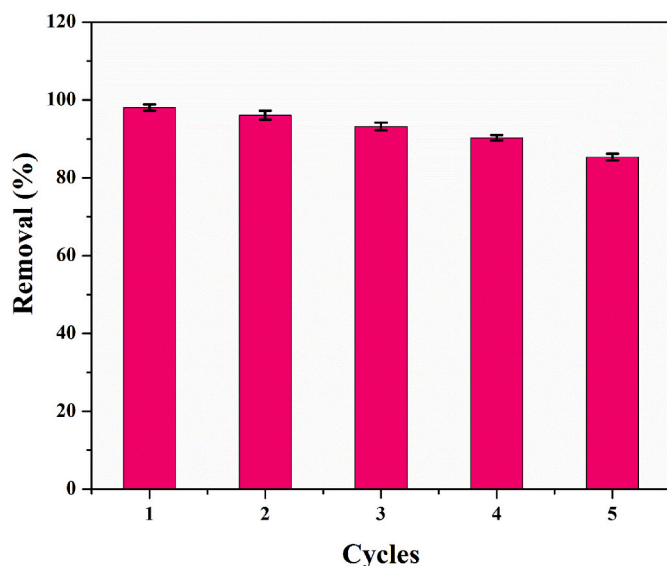


Fig. 9. Reusability of SiO₂ NPs for the removal of TC.

wastewater treatment.

3.5. Design optimization

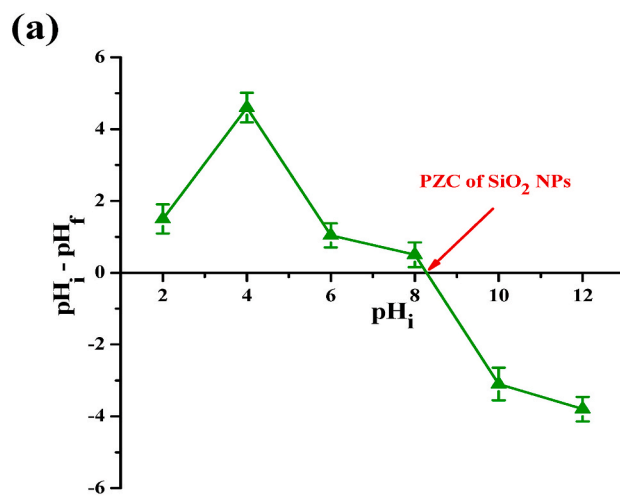
Table 5 presents the results of ANOVA analysis of the statistical significance. The F-value (162.55) and p-value (<0.0001) indicate the polynomial equation was significant for the removal of TC on SiO₂ NPs within 95% (Hu et al., 2021). The high values of R² (0.9925), adjusted R² (0.9864), and predicted R² (0.9449) indicated the excellent fit of this model to the factors selected (Aziz et al., 2021). The value of adequate precision of 37.9356 (>4) shows a high level of statistical significance (Dalia Allouss et al., 2019). The predicted TC removal using SiO₂ NPs was obtained by the developed model represented below in Eq. (10) (Dalia Allouss et al., 2019).

$$R(\%) = 98.57 - 6.84A + 3.78B + 0.8861C + 3.72AB + 0.6362AC - 0.0713BC - 3.57A^2 - 2.36B^2 - 0.6527C^2 \quad (10)$$

The residual vs. predicted and 3D response surface plots of TC percentage are illustrated in Fig. 7 and Fig. 8, respectively. The TC removal was experimentally 99.56% under optimal conditions (SiO₂ NPs dose = 0.25 g/L, C₀ = 25 mg/L, and pH = 5 at 23 ± 1 °C for 40 min) RSM-BBD modelling.

3.6. Reusability of SiO₂ NPs

To assess the applicability of the adsorbent in the full-scale operation, the reusable capacity of SiO₂ NPs was studied for the removal of TC in optimal conditions (SiO₂ NPs dose = 0.25 g/L, t = 40 min, C₀ = 50 mg/L, pH (TC) = 5, and T = 23 ± 1 °C). Therefore, the regeneration studies for evaluating the adsorption efficiency of the nanocomposite were conducted within five successive cycles using 0.1 M NaOH. According to Fig. 9, less than a 13% drop in the removal efficiency of TC occurred at the end of the fifth run, signifying the desirable reusability potential of the synthesized nanoparticles within successive runs of operation (Yang et al., 2020). This decrease was attributed to the occupation of available sites on the SiO₂ NPs surface (El Messaoudi et al., 2021b). In conclusion, the present adsorbent can be regarded as a promising material for practical application in environmental protection due to its excellent adsorption activities and high stability.



(b)

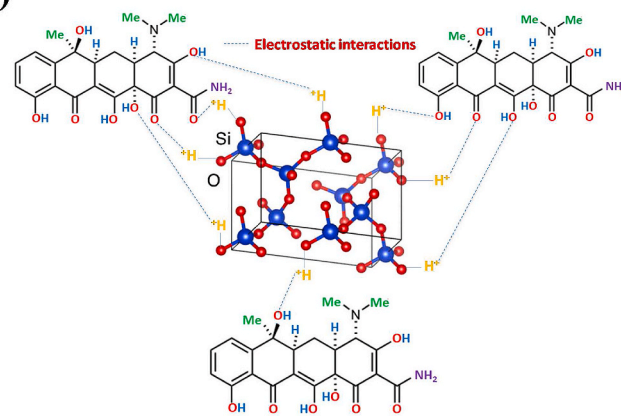


Fig. 10. (a) PZC of SiO₂ NPs and (b) Proposed adsorption mechanism of TC on SiO₂ NPs.

3.7. Proposed adsorption mechanism

Various mechanisms are involved in the adsorption of the organic compound to nanoparticle adsorbents. It was reported that both electrostatic and dispersive interaction between adsorbent and adsorbate is important in the adsorption process (Gao et al., 2019). Regarding the properties of organic adsorbate and adsorbent, the importance of each interaction is determined. As Fig. 10(a) shows, PZC of SiO₂ NPs was 8.2. The dominance of charge positive of SiO₂ NPs at pH < PZC and negatively charged when pH > PZC. The high value of PZC indicates the favourable and maximum adsorption in an acidic medium. This result confirmed the effect of pH on adsorption. The adsorption mechanism of TC on SiO₂ NPs is schematized in Fig. 10(b) based on PZC. The charge positive of SiO₂ NPs and the negative charge of TC indicates the governance of electrostatic interactions between TC and SiO₂ NPs (Gao et al., 2012).

4. Conclusions

The SiO₂ nanoparticles biosynthesized from the extract *Nerium oleander* leaves successfully with an effective method to remove tetracycline (TC) from an aqueous solution. The result of SEM-EDX, FTIR, TEM, and XRD characterization confirmed the biosynthesis of SiO₂ nanoparticles with spherical and crystallite in their structure. The parameters of the adsorption process were optimized with the variation in values of pH solution, SiO₂ NPs dose, adsorption process time, initial TC

concentration, and ionic strength. Under conditions (SiO_2 NPs dose = 0.25 g/L, $t = 40$ min, $C_0 = 50$ mg/L, pH = 5, and $T = 23 \pm 1$ °C), the TC removal was 98.62%. The kinetics and isotherm of TC adsorption on SiO_2 NPs were described as the PSO and Langmuir models, respectively. The Q_m was 552.48 mg/g. Optimization is an effective approach for modelling the sorption process of the TC on SiO_2 NPs using BBD-RSM. The recyclability study demonstrated that the SiO_2 NPs exhibited excellent reusability for TC removal. These results confirm that the SiO_2 NPs nanoparticles are suitable for removing antibiotics from wastewaters.

Credit authors statement

Noureddine El Messaoudi: Investigation, Writing – original draft, Mohammed El Khomri: Verification, Methodology, El-houssaine Ablouh: Verification, Validation, Methodology, Amal Bouich: Visualization, Software, Abdellah Lacherai: Review and editing, Conceptualization, Supervision, Amane Jada: Review and editing, Conceptualization, Supervision, Faroq Sher: Review and editing, Conceptualization, Supervision, Eder C. Lima: Review and editing, Conceptualization, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

The authors are grateful for the financial supports from the Foundation for Research Support of the State of Rio Grande do Sul – FAPERGS [19/2551-0001865-7] and National Council for Scientific and Technological Development – CNPq [303.622/2017–2] and the Engineering and Physical Sciences Research Council (EPSRC) UK.

References

- Ahamad, T., Ruksana, Chaudhary A.A., Naushad, M., Alshehri, S.M., 2019. Fabrication of MnFe₂O₄ nanoparticles embedded chitosan-diphenylureaformaldehyde resin for the removal of tetracycline from aqueous solution. *Int. J. Biol. Macromol.* 134, 180–188. <https://doi.org/10.1016/j.IJBIOMAC.2019.04.204>.
- Allouss, Dalia, Essamlali, Younes, Amadine, Othmane, Chakir, Achraf, Mohamed, Zahouily, 2019. Response surface methodology for optimization of methylene blue adsorption onto carboxymethyl cellulose-based hydrogel beads: adsorption kinetics, isotherm, thermodynamics and reusability studies. *RSC Adv.* 9, 37858–37869. <https://doi.org/10.1039/C9RA06450H>.
- Aziz, K., Aziz, F., Mamouni, R., Aziz, L., Saffaj, N., 2021. Engineering of highly Brachycton populneus shells@polyaniline bio-sorbent for efficient removal of pesticides from wastewater: optimization using BBD-RSM approach. *J. Mol. Liq.* 117092. <https://doi.org/10.1016/j.MOLLIQ.2021.117092>.
- Belkheiri, D., Fourcade, F., Geneste, F., Floner, D., Ait-Amar, H., Amrane, A., 2011. Feasibility of an electrochemical pre-treatment prior to a biological treatment for tetracycline removal. *Separ. Purif. Technol.* 83, 151–156. <https://doi.org/10.1016/J.SEPUR.2011.09.029>.
- Bembibre, A., Benamara, M., Hhiri, M., Gómez, E., Alamri, H.R., Dhahri, R., Serrà, A., 2022. Visible-light driven sonophotocatalytic removal of tetracycline using Cd-doped ZnO nanoparticles. *Chem. Eng. J.* 427, 132006. <https://doi.org/10.1016/J.CEJ.2021.132006>.
- Bentahar, S., Dbik, A., Khomri, M.E., El Messaoudi, N., Lacherai, A., 2017. Adsorption of methylene blue, crystal violet and Congo red from binary and ternary systems with natural clay: kinetic, isotherm, and thermodynamic. *J. Environ. Chem. Eng.* 5, 5921–5932. <https://doi.org/10.1016/j.jece.2017.11.003>.
- Bentahar, S., Dbik, A., Khomri, M.E., El Messaoudi, N., Lacherai, A., 2018. Removal of a cationic dye from aqueous solution by natural clay. *Groundw. Sustain. Dev.* 6, 255–262. <https://doi.org/10.1016/j.gsd.2018.02.002>.
- Chang, P.H., Li, Z., Jean, J.S., Jiang, W.T., Wang, C.J., Lin, K.H., 2012. Adsorption of tetracycline on 2:1 layered non-swelling clay mineral illite. *Appl. Clay Sci.* 67–68, 158–163. <https://doi.org/10.1016/J.CLAY.2011.11.004>.
- Daghrir, R., Drogui, P., 2013. Tetracycline antibiotics in the environment: a review. *Environ. Chem. Lett.* 113 (11), 209–227. <https://doi.org/10.1007/S10311-013-0404-8>, 2013.
- Das, S., Chakraborty, J., Chatterjee, S., Kumar, H., 2018. Prospects of biosynthesized nanomaterials for the remediation of organic and inorganic environmental contaminants. *Environ. Sci. Nano* 5, 2784–2808. <https://doi.org/10.1039/C8EN00799C>.
- Debnath, B., Majumdar, M., Bhowmik, M., Bhowmik, K.L., Debnath, A., Roy, D.N., 2020. The effective adsorption of tetracycline onto zirconia nanoparticles synthesized by novel microbial green technology. *J. Environ. Manag.* 261, 110235. <https://doi.org/10.1016/j.jenvman.2020.110235>.
- Dobrucka, R., Długasewska, J., 2016. Biosynthesis and antibacterial activity of ZnO nanoparticles using Trifolium pratense flower extract. *Saudi J. Biol. Sci.* 23, 517–523. <https://doi.org/10.1016/J.SJBS.2015.05.016>.
- Dubey, R.S., Rajesh, Y.B.R.D., More, M.A., 2015. Synthesis and characterization of SiO₂ nanoparticles via sol-gel method for industrial applications. *Mater. Today Proc.* 2, 3575–3579. <https://doi.org/10.1016/J.MATPR.2015.07.098>.
- El Khomri, M., El Messaoudi, N., Dbik, A., Bentahar, S., Lacherai, A., 2020. Efficient adsorbent derived from Argania Spinosa for the adsorption of cationic dye: kinetics, mechanism, isotherm and thermodynamic study. *Surfaces and Interfaces* 20, 100601. <https://doi.org/10.1016/j.surfin.2020.100601>.
- El Messaoudi, N., El Khomri, M., Bentahar, S., Dbik, A., Lacherai, A., Bakiz, B., 2016a. Evaluation of performance of chemically treated date stones: application for the removal of cationic dyes from aqueous solutions. *J. Taiwan Inst. Chem. Eng.* 67, 244–253. <https://doi.org/10.1016/j.jtice.2016.07.024>.
- El Messaoudi, N., El Khomri, M., Dbik, A., Bentahar, S., Lacherai, A., Bakiz, B., 2016b. Biosorption of Congo red in a fixed-bed column from aqueous solution using jujube shell: experimental and mathematical modeling. *J. Environ. Chem. Eng.* 4, 3848–3855. <https://doi.org/10.1016/j.jece.2016.08.027>.
- El Messaoudi, N., Dbik, A., El Khomri, M., Sabour, A., Bentahar, S., Lacherai, A., 2017. Date stones of Phoenix dactylifera and jujube shells of Ziziphus lotus as potential biosorbents for anionic dye removal. *Int. J. Phytoremediation* 19, 1047–1052. <https://doi.org/10.1080/15226514.2017.1319331>.
- El Messaoudi, N., El Khomri, M., Chlif, N., Chegini, Z.G., Dbik, A., Bentahar, S., Lacherai, A., 2021a. Desorption of Congo red from dye-loaded Phoenix dactylifera date stones and Ziziphus lotus jujube shells. *Groundw. Sustain. Dev.* 12, 100552. <https://doi.org/10.1016/j.gsd.2021.100552>.
- El Messaoudi, N., El Khomri, M., Dabagh, A., Chegini, Z.G., Dbik, A., Bentahar, S., Lacherai, A., Iqbal, M., Jada, A., Sher, F., Lima, E.C., 2021b. Synthesis of a novel nanocomposite based on date stones/CuFe₂O₄ nanoparticles for eliminating cationic and anionic dyes from aqueous solution. *Int. J. Environ. Stud.* 1–19. <https://doi.org/10.1080/00207233.2021.1929469>.
- Epps, A. Van, Blaney, L., 2016. Antibiotic residues in animal waste: occurrence and degradation in conventional agricultural waste management practices. *Curr. Pollut. Reports* 23 (2), 135–155. <https://doi.org/10.1007/S40726-016-0037-1>, 2016.
- Fiol, N., Villacueva, I., 2009. Determination of sorbent point zero charge: usefulness in sorption studies. *Environ. Chem. Lett.* 7, 79–84. <https://doi.org/10.1007/s10311-008-0139-0>.
- Freundlich, H., 1907. Über die Adsorption in Lösungen. *Zeitschrift für Phys. Chemie* 57U 385–470. <https://doi.org/10.1515/ZPCH-1907-5723>.
- Fu, Y., Peng, L., Zeng, Q., Yang, Y., Song, H., Shao, J., Liu, S., Gu, J., 2015. High efficient removal of tetracycline from solution by degradation and flocculation with nanoscale zerovalent iron. *Chem. Eng. J.* 270, 631–640. <https://doi.org/10.1016/J.CEJ.2015.02.070>.
- Gao, Y., Li, Y., Zhang, L., Huang, H., Hu, J., Shah, S.M., Su, X., 2012. Adsorption and removal of tetracycline antibiotics from aqueous solution by graphene oxide. *J. Colloid Interface Sci.* 368, 540–546. <https://doi.org/10.1016/J.JCIS.2011.11.015>.
- Gao, B., Li, P., Yang, R., Li, A., Yang, H., 2019. Investigation of multiple adsorption mechanisms for efficient removal of ofloxacin from water using lignin-based adsorbents. *Sci. Rep.* 9 (1), 1–13. <https://doi.org/10.1038/s41598-018-37206-1>, 2019.
- Graaf, G.H., Scholtens, H., Stamhuis, E.J., Beenackers, A.A.C.M., 1990. Intra-particle diffusion limitations in low-pressure methanol synthesis. *Chem. Eng. Sci.* 45, 773–783. [https://doi.org/10.1016/0009-2509\(90\)85001-T](https://doi.org/10.1016/0009-2509(90)85001-T).
- Grenni, P., Ancona, V., Barra Caracciolo, A., 2018. Ecological effects of antibiotics on natural ecosystems: a review. *Microchem. J.* 136, 25–39. <https://doi.org/10.1016/J.MICROC.2017.02.006>.
- Hao, D., Chen, Y., Zhang, Y., You, N., 2021. Nanocomposites of zero-valent iron@biochar derived from agricultural wastes for adsorptive removal of tetracyclines. *Chemosphere* 284, 131342. <https://doi.org/10.1016/J.CHEMOSPHERE.2021.131342>.
- Ho, Y.S., McKay, G., 1998. Sorption of dye from aqueous solution by peat. *Chem. Eng. J.* 70, 115–124. [https://doi.org/10.1016/S0923-0467\(98\)00076-1](https://doi.org/10.1016/S0923-0467(98)00076-1).
- Hosseini, S.A., Vossoughi, M., Mahmoodi, N.M., Sadrzadeh, M., 2018. Efficient dye removal from aqueous solution by high-performance electrospun nanofibrous membranes through incorporation of SiO₂ nanoparticles. *J. Clean. Prod.* 183, 1197–1206. <https://doi.org/10.1016/j.jclepro.2018.02.168>.
- Hu, Y., ying, Pan, C., Zheng, X., Hu, F., Xu, L., Xu, G., Jian, Y., Peng, X., 2021. Prediction and optimization of adsorption properties for Cs+ on NiSiO₄@NiAlFe LDHs hollow spheres from aqueous solution: kinetics, isotherms, and BBD model. *J. Hazard Mater.* 401, 123374. <https://doi.org/10.1016/J.JHAZMAT.2020.123374>.
- Jawad, A.H., Malek, N.N.A., Abdulhameed, A.S., Razuan, R., 2020. Synthesis of magnetic chitosan-fly ash/Fe₃O₄ composite for adsorption of reactive orange 16 dye: optimization by box-behnken design. *J. Polym. Environ.* 28, 1068–1082. <https://doi.org/10.1007/s10924-020-01669-z>.
- Jin, J., Yang, Z., Xiong, W., Zhou, Y., Xu, R., Zhang, Y., Cao, J., Li, X., Zhou, C., 2019. Cu and Co nanoparticles co-doped MIL-101 as a novel adsorbent for efficient removal of tetracycline from aqueous solutions. *Sci. Total Environ.* 650, 408–418. <https://doi.org/10.1016/J.SCITOTENV.2018.08.434>.

- Johnson, R.D., Arnold, F.H., 1995. The temkin isotherm describes heterogeneous protein adsorption. *Biochim. Biophys. Acta Protein Struct. Mol. Enzymol.* 1247, 293–297. [https://doi.org/10.1016/0167-4838\(95\)00006-G](https://doi.org/10.1016/0167-4838(95)00006-G).
- Langmuir, I., 1918. The adsorption of gases on plane surfaces of glass, mica and platinum. *J. Am. Chem. Soc.* 40, 1361–1403. <https://doi.org/10.1021/JA02242A004>.
- Li, Z., Schulz, L., Ackley, C., Fenske, N., 2010. Adsorption of tetracycline on kaolinite with pH-dependent surface charges. *J. Colloid Interface Sci.* 351, 254–260. <https://doi.org/10.1016/J.JCIS.2010.07.034>.
- Li, S., Huang, W., Yang, P., Li, Z., Xia, B., Li, M., Xue, C., Liu, D., 2021. One-pot synthesis of N-doped carbon intercalated molybdenum disulfide nanohybrid for enhanced adsorption of tetracycline from aqueous solutions. *Sci. Total Environ.* 754, 141925. <https://doi.org/10.1016/J.SCTOTENV.2020.141925>.
- Liu, F., Zhang, W., Chen, W., Wang, Jing, Yang, Q., Zhu, W., Wang, Jianlong, 2017. One-pot synthesis of NiFe₂O₄ integrated with EDTA-derived carbon dots for enhanced removal of tetracycline. *Chem. Eng. J.* 310, 187–196. <https://doi.org/10.1016/J.CEJ.2016.10.116>.
- Maged, A., Iqbal, J., Kharbush, S., Ismael, I.S., Bhatnagar, A., 2020. Tuning tetracycline removal from aqueous solution onto activated 2:1 layered clay mineral: characterization, sorption and mechanistic studies. *J. Hazard Mater.* 384, 121320. <https://doi.org/10.1016/j.jhazmat.2019.121320>.
- Martín, A., Caldelas, C., Weiss, D., Aranjuelo, I., Navarro, E., 2018. Assessment of metal immission in urban environments using elemental concentrations and zinc isotope signatures in leaves of Nerium oleander. *Environ. Sci. Technol.* 52, 2071–2080. <https://doi.org/10.1021/ACS.EST.7B00617>.
- Masjedi-Arani, M., Ghanbari, D., Salavati-Niasari, M., Bagheri, S., 2015. Sonochemical synthesis of spherical silica nanoparticles and polymeric nanocomposites. *J. Cluster Sci.* 271 (27), 39–53. <https://doi.org/10.1007/S10876-015-0897-3>, 2015.
- Mohammed, A.A., Kareem, S.L., 2019. Adsorption of tetracycline from wastewater by using Pistachio shell coated with ZnO nanoparticles: equilibrium, kinetic and isotherm studies. *Alexandria Eng. J.* 58, 917–928. <https://doi.org/10.1016/J.AEJ.2019.08.006>.
- Mohammed, A.A., Al-Musawi, T.J., Kareem, S.L., Zarrabi, M., Al-Ma'abreh, A.M., 2020. Simultaneous adsorption of tetracycline, amoxicillin, and ciprofloxacin by pistachio shell powder coated with zinc oxide nanoparticles. *Arab. J. Chem.* 13, 4629–4643. <https://doi.org/10.1016/J.ARABJC.2019.10.010>.
- Niksefat, N., Jahanshahi, M., Rahimpour, A., 2014. The effect of SiO₂ nanoparticles on morphology and performance of thin film composite membranes for forward osmosis application. *Desalination* 343, 140–146. <https://doi.org/10.1016/J.DESAL.2014.03.031>.
- Nita, C., Fullenwarth, J., Monconduit, L., Le Meins, J.M., Fioux, P., Parmentier, J., Matei Ghimbeu, C., 2019. Eco-friendly synthesis of SiO₂ nanoparticles confined in hard carbon: a promising material with unexpected mechanism for Li-ion batteries. *Carbon* N. Y. 143, 598–609. <https://doi.org/10.1016/J.CARBON.2018.11.069>.
- Nurhasanah, I., Kadarisman, Gunawan V., Sutanto, H., 2020. Cerium oxide nanoparticles application for rapid adsorptive removal of tetracycline in water. *J. Environ. Chem. Eng.* 8, 103613. <https://doi.org/10.1016/J.JECE.2019.103613>.
- Parolo, M.E., Savini, M.C., Vallés, J.M., Baschini, M.T., Avena, M.J., 2008. Tetracycline adsorption on montmorillonite: pH and ionic strength effects. *Appl. Clay Sci.* 40, 179–186. <https://doi.org/10.1016/J.CLAY.2007.08.003>.
- Potapov, V., Fediuk, R., Gorev, D., 2020. Membrane concentration of hydrothermal SiO₂ nanoparticles. *Separ. Purif. Technol.* 251, 117290. <https://doi.org/10.1016/J.SEPUR.2020.117290>.
- Rafiq, S.M., Heydarinasab, A., 2017. Mesoporous chitosan-SiO₂ nanoparticles: synthesis, characterization, and CO₂ adsorption capacity. *ACS Sustain. Chem. Eng.* 5, 10379–10386. <https://doi.org/10.1021/ACSSUSCHEMENG.7B02388>.
- Rashidi Nodeh, H., Sereshti, H., Beirakbadi, E., Razmkhah, K., 2020. Synthesis and application of lanthanum sulfide nanoparticles for removal of tetracycline from aqueous media. *Int. J. Environ. Sci. Technol.* 17, 819–828. <https://doi.org/10.1007/s13762-019-02399-z>.
- Ravikumar, K.V.G., Sudakaran, S.V., Ravichandran, K., Pulimi, M., Natarajan, C., Mukherjee, A., 2019. Green synthesis of NiFe nano particles using Punica granatum peel extract for tetracycline removal. *J. Clean. Prod.* 210, 767–776. <https://doi.org/10.1016/j.jclepro.2018.11.108>.
- Rostam, A.B., Taghizadeh, M., 2020. Advanced oxidation processes integrated by membrane reactors and bioreactors for various wastewater treatments: a critical review. *J. Environ. Chem. Eng.* 8, 104566. <https://doi.org/10.1016/J.JECE.2020.104566>.
- Scaria, J., Anupama, K.V., Nidheesh, P.V., 2021. Tetracyclines in the environment: an overview on the occurrence, fate, toxicity, detection, removal methods, and sludge management. *Sci. Total Environ.* 771, 145291. <https://doi.org/10.1016/J.SCTOTENV.2021.145291>.
- Sebeia, N., Jabli, M., Ghith, A., 2019. Biological synthesis of copper nanoparticles, using Nerium oleander leaves extract: characterization and study of their interaction with organic dyes. *Inorg. Chem. Commun.* 105, 36–46. <https://doi.org/10.1016/J.INOCHE.2019.04.023>.
- Shaligram, N.S., Bule, M., Bhambure, R., Singhal, R.S., Singh, S.K., Szakacs, G., Pandey, A., 2009. Biosynthesis of silver nanoparticles using aqueous extract from the compactin producing fungal strain. *Process Biochem.* 44, 939–943. <https://doi.org/10.1016/J.PROCBIO.2009.04.009>.
- Sharma, G., Bhogal, S., Kumar, A., Naushad, M., Sharma, S., Ahamad, T., Stadler, F.J., 2020. AgO/MgO/FeO@Si₃N₄ nanocomposite with robust adsorption capacity for tetracycline antibiotic removal from aqueous system. *Adv. Powder Technol.* 31, 4310–4318. <https://doi.org/10.1016/J.APT.2020.09.006>.
- Sharma, P., Kherb, J., Prakash, J., Kaushal, R., 2021. A novel and facile green synthesis of SiO₂ nanoparticles for removal of toxic water pollutants. *Appl. Nanosci.* 2021, 1–13. <https://doi.org/10.1007/S13204-021-01898-1>.
- Simonin, J.P., 2016. On the comparison of pseudo-first order and pseudo-second order rate laws in the modeling of adsorption kinetics. *Chem. Eng. J.* 300, 254–263. <https://doi.org/10.1016/J.CEJ.2016.04.079>.
- Sodhi, K.K., Kumar, M., Balan, B., Dhulaniya, A.S., Shree, P., Sharma, N., Singh, D.K., 2021. Perspectives on the antibiotic contamination, resistance, metabolomics, and systemic remediation. *SN Appl. Sci.* 32 (3), 1–25. <https://doi.org/10.1007/S42452-020-04003-3>, 2021.
- Song, Y.X., Chen, S., You, N., Fan, H.T., Sun, L.N., 2020. Nanocomposites of zero-valent Iron@Activated carbon derived from corn stalk for adsorptive removal of tetracycline antibiotics. *Chemosphere* 255, 126917. <https://doi.org/10.1016/J.CHEMOSPHERE.2020.126917>.
- Ullah, Z., Man, Z., Khan, A.S., Muhammad, N., Mahmood, H., Ben Ghanem, O., Ahmad, P., Hassan Shah, M.U., Mamoon-Ur-Rashid, Raheel, M., 2019. Extraction of valuable chemicals from sustainable rice husk waste using ultrasonic assisted ionic liquids technology. *J. Clean. Prod.* 220, 620–629. <https://doi.org/10.1016/J.JCLEPRO.2019.02.041>.
- Van, T.T.H., Yidana, Z., Smooker, P.M., Coloe, P.J., 2020. Antibiotic use in food animals worldwide, with a focus on Africa: pluses and minuses. *J. Glob. Antimicrob. Resist.* 20, 170–177. <https://doi.org/10.1016/J.JGAR.2019.07.031>.
- Wang, X., Li, F., Hu, X., Hua, T., 2021. Electrochemical advanced oxidation processes coupled with membrane filtration for degrading antibiotic residues: a review on its potential applications, advances, and challenges. *Sci. Total Environ.* 784, 146912. <https://doi.org/10.1016/J.SCTOTENV.2021.146912>.
- Yang, S., Hu, J., Chen, C., Shao, D., Wang, X., 2011. Mutual effects of Pb(II) and humic acid adsorption on multiwalled carbon nanotubes/polyacrylamide composites from aqueous solutions. *Environ. Sci. Technol.* 45, 3621–3627. <https://doi.org/10.1021/ES104047D>.
- Yang, G., Gao, Q., Yang, S., Yin, S., Cai, X., Yu, X., Zhang, S., Fang, Y., 2020. Strong adsorption of tetracycline hydrochloride on magnetic carbon-coated cobalt oxide nanoparticles. *Chemosphere* 239, 124831. <https://doi.org/10.1016/j.chemosphere.2019.124831>.
- Yu, F., Li, Y., Han, S., Ma, J., 2016. Adsorptive removal of antibiotics from aqueous solution using carbon materials. *Chemosphere* 153, 365–385. <https://doi.org/10.1016/J.CHEMOSPHERE.2016.03.083>.
- Yue, Y., Peng, Z., Wang, W., Cai, Y., Tan, F., Wang, X., Qiao, X., 2019. Facile preparation of MgO-loaded SiO₂ nanocomposites for tetracycline removal from aqueous solution. *Powder Technol.* 347, 1–9. <https://doi.org/10.1016/J.POWTEC.2019.02.034>.
- Zhang, X., Lin, X., He, Y., Chen, Y., Luo, X., Shang, R., 2019. Study on adsorption of tetracycline by Cu-immobilized alginate adsorbent from water environment. *Int. J. Biol. Macromol.* 124, 418–428. <https://doi.org/10.1016/J.IJBIOMAC.2018.11.218>.
- Zhou, Y., Liu, X., Xiang, Y., Wang, P., Zhang, J., Zhang, F., Wei, J., Luo, L., Lei, M., Tang, L., 2017. Modification of biochar derived from sawdust and its application in removal of tetracycline and copper from aqueous solution: adsorption mechanism and modelling. *Bioreour. Technol.* 245, 266–273. <https://doi.org/10.1016/J.BIORTECH.2017.08.178>.
- Zhou, J., Ma, F., Guo, H., 2020. Adsorption behavior of tetracycline from aqueous solution on ferroferric oxide nanoparticles assisted powdered activated carbon. *Chem. Eng. J.* 384, 123290. <https://doi.org/10.1016/J.CEJ.2019.123290>.
- Zhu, X.D., Wang, Y.J., Sun, R.J., Zhou, D.M., 2013. Photocatalytic degradation of tetracycline in aqueous solution by nanosized TiO₂. *Chemosphere* 92, 925–932. <https://doi.org/10.1016/J.CHEMOSPHERE.2013.02.066>.